

Answer Key & Teacher Notes

Session 1 – Area of Triangles • Session 2 – Missing Measures in Triangles

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Area of Triangles

TALK PROMPTS

- Why does the triangle formula include a half? Explain with two copies of the triangle.
- Why does halving the product give the triangle and not the parallelogram?

WATCH FOR

- Forgetting to take half (finding the parallelogram's area).
- Using a slanted side as the height.
- Not seeing that the legs of a right triangle are the base and height.

PARTNER TASK

Cut Sandwich and Right-Triangle Sail. One partner cuts a paper rectangle on a diagonal and shows the two congruent triangles; the other writes $A = \frac{1}{2} \times b \times h$ and solves. Switch roles with Recess Wedge: base 14 ft, height 9 ft (63 ft²).

SESSION 2 • TEACHER NOTES

Missing Measures in Triangles

TALK PROMPTS

- How is finding a missing base different from finding an area? What changes?
- What would change in your steps if the height were given instead of the base?

WATCH FOR

- Dividing the area by one measure without doubling ($66 \div 11 = 6$).
- Writing a length answer in square units (12 ft²).
- Halving the area instead of doubling it to undo the $\frac{1}{2}$.

PARTNER TASK

Quilt Piece and Roof Gable. One partner substitutes into $A = \frac{1}{2} \times b \times h$; the other doubles the area, divides, and checks by working forward. Switch roles with Awning Panel.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.G.A.1	1	2	1	1	1	2	2	3	3	1	1
S1 • Form B	6.G.A.1	1	2	1	1	1	2	2	3	3	1	1
S2 • Form A	6.G.A.1	1	2	1	1	1	2	2	3	3	1	1
S2 • Form B	6.G.A.1	1	2	1	1	1	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

right triangle: A triangle with one **square** corner.

height: The distance from the base to the opposite corner, at a **right** angle.

congruent: Exactly the same **size** and shape.

- Two congruent triangles make a **parallelogram**.
- So a triangle's area is **half** of base $\times$ height: $A = \frac{1}{2} \times b \times h$.
- In a right triangle, the two **legs** are the base and the height.

Example: $14 \times 9 = 126$ 126 cm^2 is the area of the parallelogram made by two triangles. $126 \div 2 = 63$
Answer: The area is **63** cm^2 .

YOU TRY

1. a. 35 m^2 b. 20 ft^2
2. $\frac{1}{2} \times 12 \times 5 = 30 \text{ cm}^2$ (the 13 cm side is not used)

EXIT TICKET

1. a. 99 in^2 b. 14 cm^2
2. $\frac{1}{2} \times 9 \times 6 = 27 \text{ in}^2$
3. **B** ($A = \frac{1}{2} \times b \times h$)

PRACTICE FORM A

1. a. 30 cm^2 b. 21 in^2 c. 70 ft^2 d. 44 m^2
2. a. $12 \times 20 = 240 \text{ in}^2$ b. $240 \div 2 = 120 \text{ in}^2$
3. a. $\frac{1}{2} \times 10 \times 8 = 40 \text{ cm}^2$ b. $2 \times 40 = 80 \text{ cm}^2$
4. $\frac{1}{2} \times 6 \times 11 = 33 \text{ ft}^2$
5. $\frac{1}{2} \times 30 \times 26 = 390 \text{ cm}^2$
6. $\frac{1}{2} \times 14 \times 9 = 63 \text{ ft}^2$
7. P: $\frac{1}{2} \times 18 \times 10 = 90 \text{ m}^2$. Q: $\frac{1}{2} \times 12 \times 13 = 78 \text{ m}^2$.
Banner P covers more.
8. a. $\frac{1}{2} \times 24 \times 15 = 180 \text{ ft}^2$ b. $180 \div 30 = 6$ bags c. $6 \times \$6 = \36 . If the base doubles, the area doubles to 360 ft^2 , so the cost doubles to $\$72$.
9. 220 cm^2 is the area of the parallelogram made by two triangles. He forgot to take half: $A = \frac{1}{2} \times 22 \times 10 = 110 \text{ cm}^2$.
10. **B** (42 in^2)
11. **C** ($1/2$)

Self-check word **HALVING**

PRACTICE FORM B

1. a. 40 cm^2 b. 18 in^2 c. 66 ft^2 d. 52 m^2
2. a. $10 \times 18 = 180 \text{ in}^2$ b. $180 \div 2 = 90 \text{ in}^2$
3. a. $\frac{1}{2} \times 12 \times 9 = 54 \text{ cm}^2$ b. $2 \times 54 = 108 \text{ cm}^2$
4. $\frac{1}{2} \times 8 \times 15 = 60 \text{ ft}^2$
5. $\frac{1}{2} \times 24 \times 21 = 252 \text{ cm}^2$
6. $\frac{1}{2} \times 16 \times 7 = 56 \text{ ft}^2$
7. P: $\frac{1}{2} \times 16 \times 9 = 72 \text{ m}^2$. Q: $\frac{1}{2} \times 12 \times 14 = 84 \text{ m}^2$.
Banner Q covers more.
8. a. $\frac{1}{2} \times 20 \times 18 = 180 \text{ ft}^2$ b. $180 \div 30 = 6$ bags c. $6 \times \$5 = \30 . If the base doubles, the area doubles to 360 ft^2 , so the cost doubles to $\$60$.
9. 140 m^2 is the area of the parallelogram made by two triangles. She forgot to take half: $A = \frac{1}{2} \times 20 \times 7 = 70 \text{ m}^2$.
10. **B** (40 m^2)
11. **A** (a square corner (right angle))

Self-check word **PRODUCT**

GUIDED NOTES

area: The space inside a shape, measured in **square** units.

formula: A rule written with **letters** and symbols.

inverse operation: An operation that **undoes** another.

- Undo the half first: **double** the area.
- Then **divide** by the measure you know.
- A base or height is a length, so use units like cm, not **cm²**.

Example: $40 = \frac{1}{2} \times 10 \times h$ Double the area: $2 \times 40 = 80$
Divide by the base: $80 \div 10 = 8$ Answer: The height is **8** cm.

YOU TRY

1. a. 10 ft b. 6 m
2. $2 \times 48 = 96$; $96 \div 16 = 6$ m

EXIT TICKET

1. a. 12 cm b. 18 in
2. $2 \times 35 = 70$; $70 \div 10 = 7$ ft
3. **B** (half as big as it should be)

PRACTICE FORM A

1. a. 9 cm b. 8 in c. 10 m d. 9 ft
2. a. $126 = \frac{1}{2} \times b \times 18$ b. $2 \times 126 = 252$; $252 \div 18 = 14$ yd
3. $2 \times 54 = 108$; $108 \div 12 = 9$ in.
4. $2 \times 117 = 234$; $234 \div 9 = 26$ ft
5. $2 \times 60 = 120$; $120 \div 15 = 8$ ft
6. No. $\frac{1}{2} \times 28 \times 7 = 98$, not 84. Base = $2 \times 84 \div 7 = 168 \div 7 = 24$ cm.
7. $2 \times 45 = 90$; $90 \div 5 = 18$ m
8. a. $2 \times 120 \div 16 = 15$ ft b. $2 \times 120 \div 12 = 20$ ft c. Many base-height pairs multiply to 240 (16×15 and 20×12), so the triangles have the same area but different shapes.
9. He skipped undoing the $\frac{1}{2}$. Double the area first: $2 \times 66 = 132$, and $132 \div 11 = 12$ ft. Check: $\frac{1}{2} \times 11 \times 12 = 66$.
10. **B** (12 cm)
11. **C** (10 in.)

Self-check word MISSING

PRACTICE FORM B

1. a. 8 cm b. 8 in c. 10 m d. 6 ft
2. a. $132 = \frac{1}{2} \times b \times 12$ b. $2 \times 132 = 264$; $264 \div 12 = 22$ yd
3. $2 \times 42 = 84$; $84 \div 7 = 12$ in.
4. $2 \times 135 = 270$; $270 \div 15 = 18$ ft
5. $2 \times 70 = 140$; $140 \div 14 = 10$ ft
6. No. $\frac{1}{2} \times 10 \times 9 = 45$, not 90. Base = $2 \times 90 \div 9 = 180 \div 9 = 20$ ft.
7. $2 \times 52 = 104$; $104 \div 8 = 13$ m
8. a. $2 \times 150 \div 20 = 15$ ft b. $2 \times 150 \div 10 = 30$ ft c. Many base-height pairs multiply to 300 (20×15 and 30×10), so the triangles have the same area but different shapes.
9. He skipped undoing the $\frac{1}{2}$. Double the area first: $2 \times 90 = 180$, and $180 \div 15 = 12$ cm. Check: $\frac{1}{2} \times 15 \times 12 = 90$.
10. **B** (8 m)
11. **C** (10 in.)

Self-check word INVERSE

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 · AREA OF TRIANGLES

1. $\frac{1}{2} \times 12 \times 7 = 42 \text{ cm}^2$
2. $\frac{1}{2} \times 9 \times 8 = 36 \text{ in}^2$
3. a. $\frac{1}{2} \times 10 \times 9 = 45 \text{ cm}^2$ b. $\frac{1}{2} \times 5 \times 11 = 27.5 \text{ ft}^2$
4. Area = $\frac{1}{2} \times 15 \times 6 = 45 \text{ ft}^2$; $45 \div 9 = 5$ bags
5. He forgot to take half (40 m^2 is the parallelogram's area). Correct area: $\frac{1}{2} \times 8 \times 5 = 20 \text{ m}^2$.

SHEET 2 · MISSING MEASURES IN TRIANGLES

1. A: $h = 2 \times 28 \div 8 = 7 \text{ in}$; B: $b = 2 \times 54 \div 9 = 12 \text{ ft}$
2. $2 \times 48 = 96$; $96 \div 16 = 6 \text{ m}$
3. a. $2 \times 40 \div 10 = 8 \text{ m}$ b. $2 \times 27 \div 6 = 9 \text{ in}$
4. $2 \times 35 = 70$; $70 \div 10 = 7 \text{ ft}$
5. Neither. $P = \frac{1}{2} \times 12 \times 5 = 30 \text{ cm}^2$ and $Q = \frac{1}{2} \times 6 \times 10 = 30 \text{ cm}^2$, so the areas are equal.
6. **C** (12 units)
7. **B** ($\frac{1}{2} \times 18 \times 5$)